What 6.8 million mammograms can teach us about breast cancer screening

What is mammographic density?

It describes how much glandular tissue the breast has relative to fat, as seen on the mammogram. Radiologists rate it in four levels. The top two count as high tissue density.

Why does it matter?

High tissue density can hide tumors on the image, and it is linked to a higher risk of breast cancer.

Our question

Can common risk factors such as age, BMI, menopause or family history predict a mammogram's tissue density?

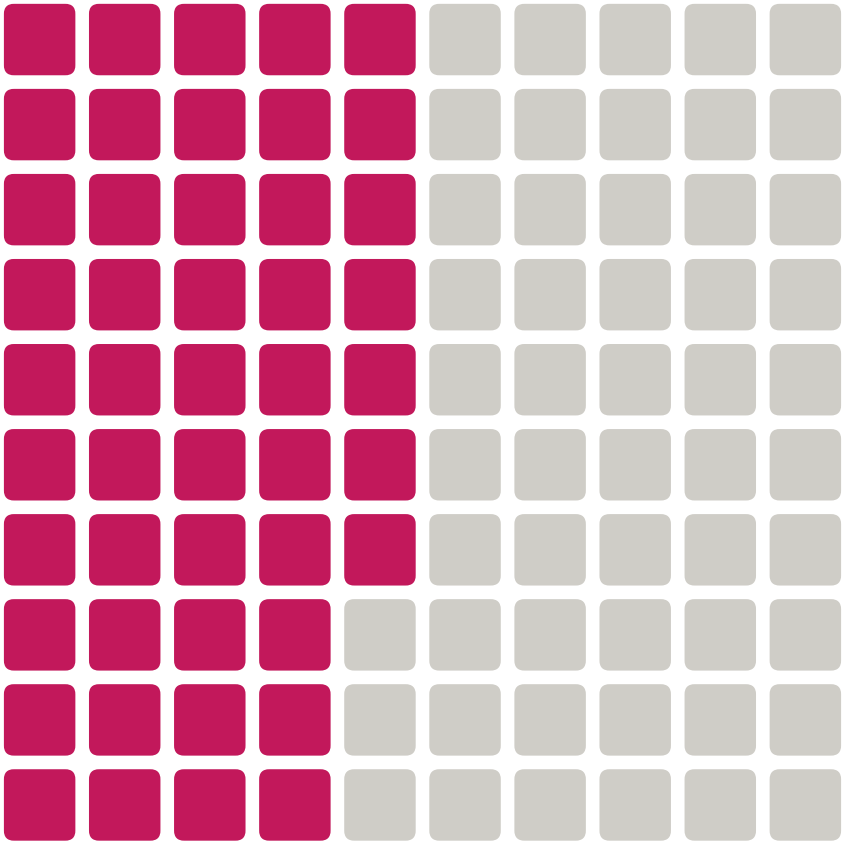
Swipe to see what the data show >

Data: Breast Cancer Surveillance Consortium, 6.8M screening mammograms (2005 to 2017)

Exploratory analysis, not medical advice

Almost half of screening mammograms show high tissue density

High density is common, so it is worth knowing whether it applies to you.



47 out of every 100 mammograms

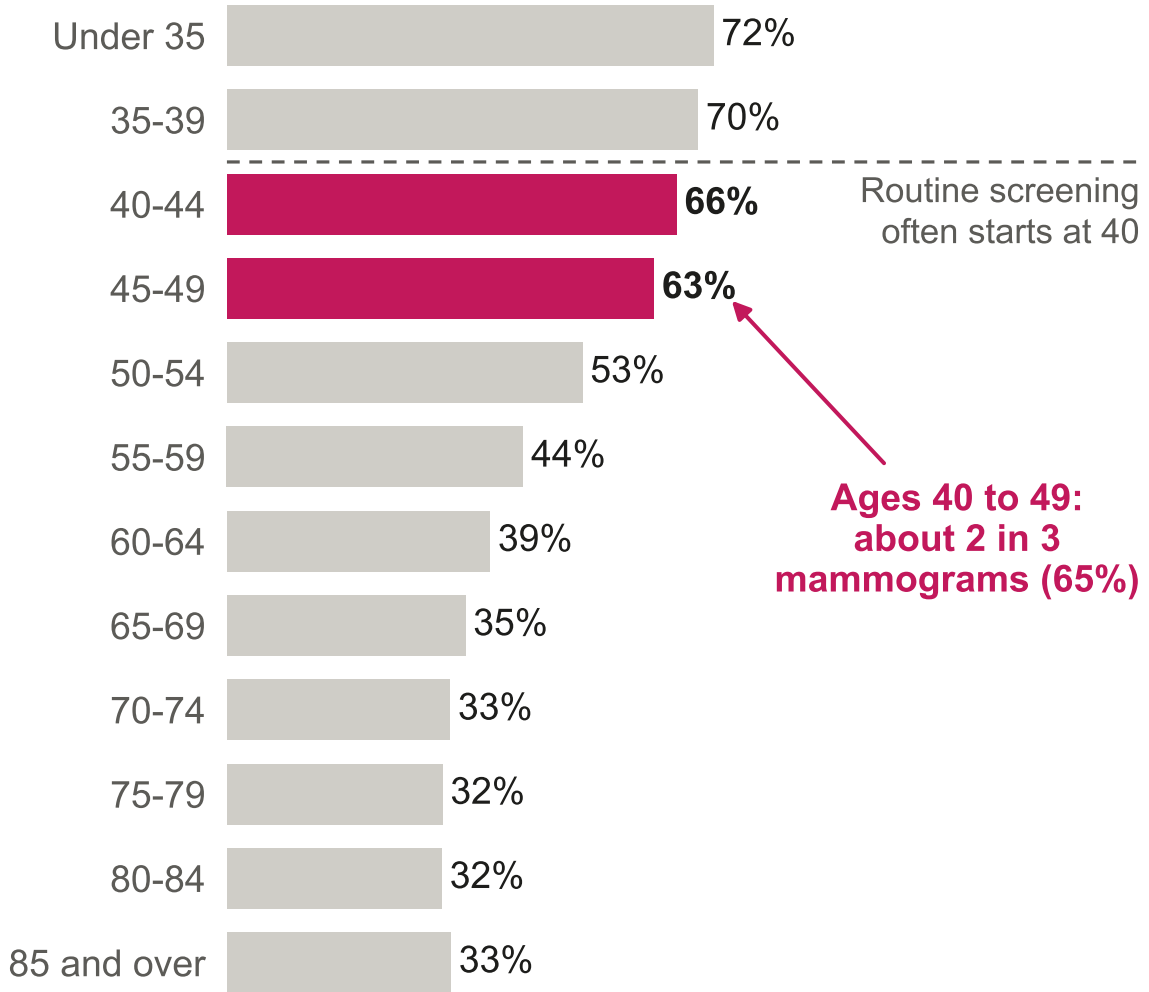
based on 6.0M mammograms with a recorded density

Data: Breast Cancer Surveillance Consortium, 6.8M screening mammograms (2005 to 2017)

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High tissue density is most common at younger ages

Routine screening often starts at 40, when high density is still common.



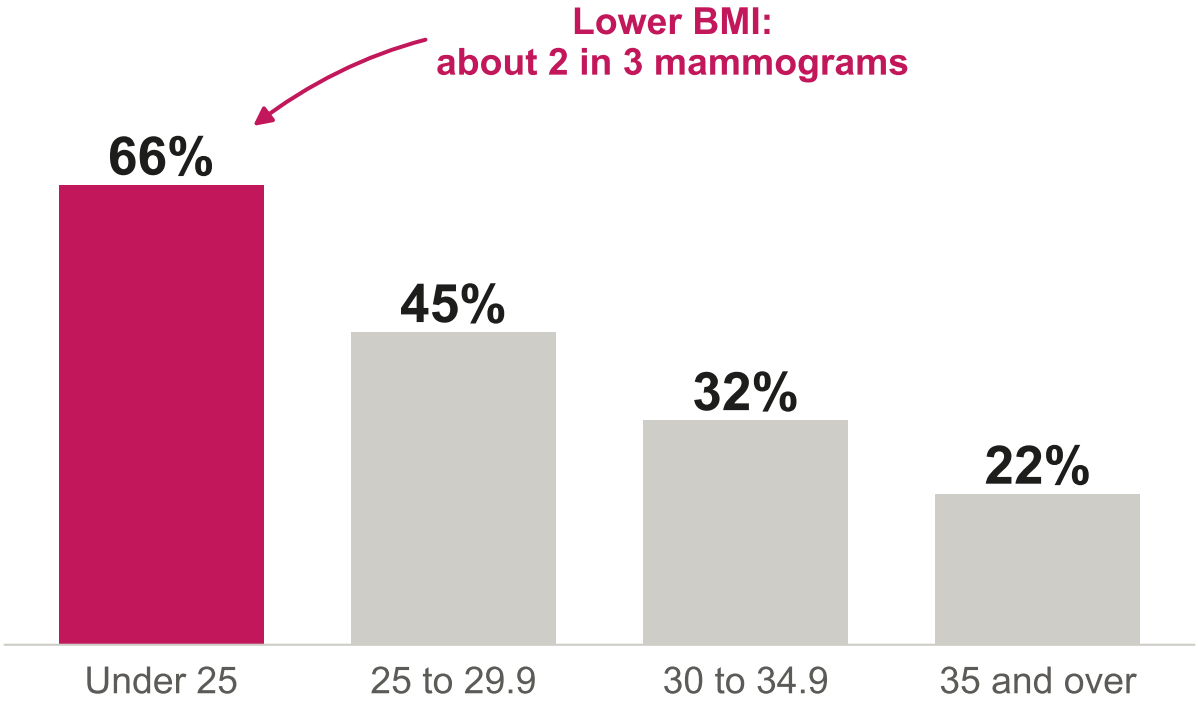
Share of screening mammograms with high tissue density, by age group

Data: Breast Cancer Surveillance Consortium, 6.8M screening mammograms (2005 to 2017)

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Mammographic density is higher at lower BMI

Tissue density shapes what a mammogram can show, at every BMI.



Share of screening mammograms with high tissue density, by BMI group

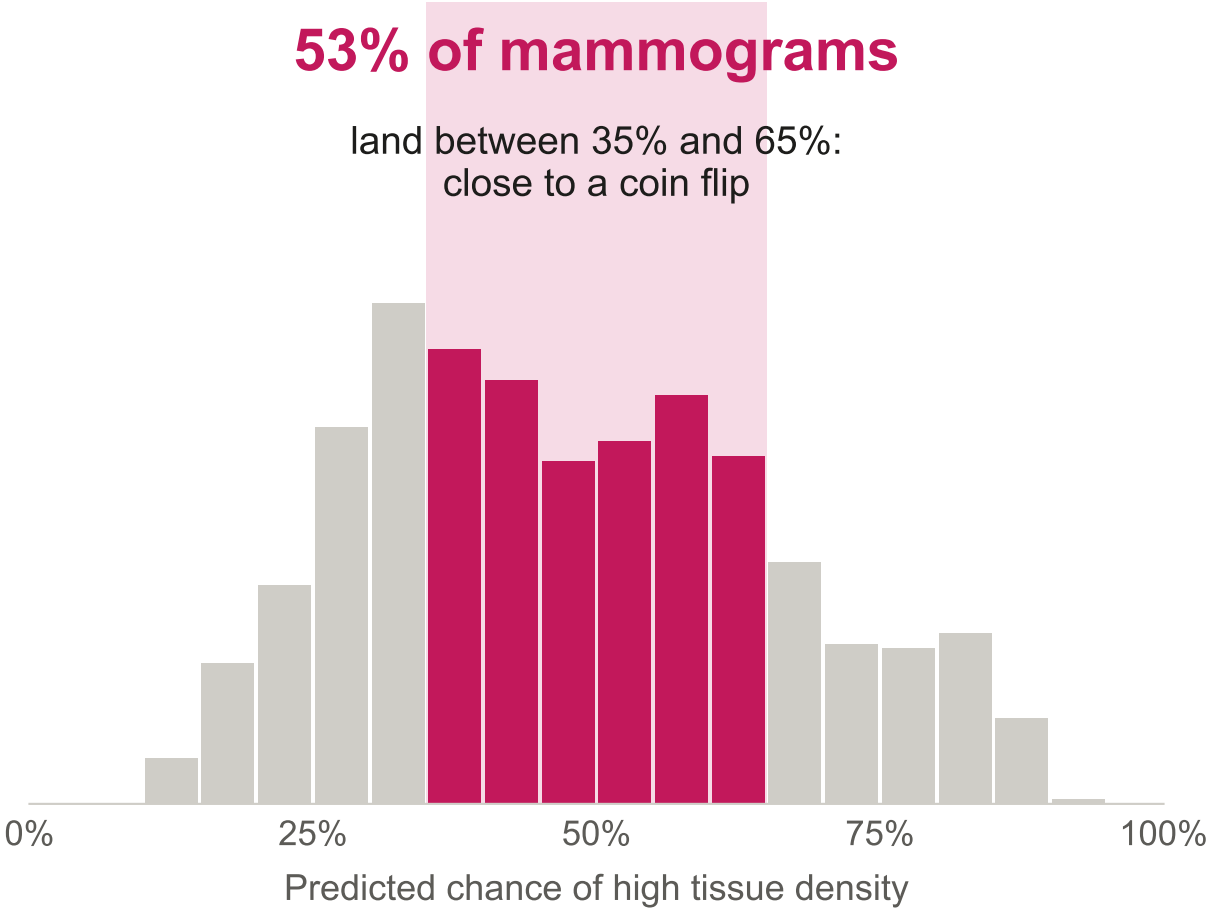
Higher BMI is still linked to higher breast cancer risk after menopause. Density is one factor among many.

Data: Breast Cancer Surveillance Consortium, 6.8M screening mammograms (2005 to 2017)

Exploratory analysis, not medical advice

Risk factors alone cannot tell us the result

Only the mammogram itself shows the tissue density, one more reason to get screened.



Only 10% get a confident prediction (below 20% or above 80%).

Predictions from 8 common risk factors, checked on 1.2M mammograms that were not used to build the predictions.

Data: Breast Cancer Surveillance Consortium, 6.8M screening mammograms (2005 to 2017)

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Family history seems unrelated to tissue density, until you compare by age

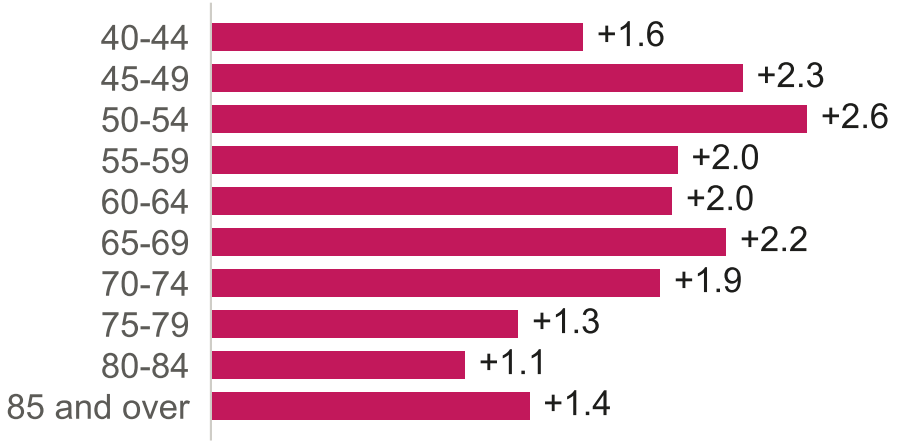
Look closer before drawing conclusions: an average can hide a real pattern.

All ages together: no difference



Within each age group: higher with a family history

Difference in percentage points, ages 40 and over



Why? Mammograms with a family history came more often from older women (49% aged 60 and over vs. 42%), and density falls with age.

Data: Breast Cancer Surveillance Consortium, 6.8M screening mammograms (2005 to 2017)

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No checklist replaces screening



Risk factors explain only part of the picture. The mammogram shows the rest.

1

Know your screening schedule

Ask when and how often screening is recommended for you.

2

Ask your doctor about your mammographic density

It is often noted in the mammogram report.

3

Ask whether supplemental screening makes sense for you

For some women with high density, extra imaging can be an option.

Talk to your doctor about what is right for you.

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